CO3-AT1
Case Study
DSA0405- FUNDAMENTALS OF DATA SCIENCE

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Case Study 1 — Employee Salaries

Data (₹ lakhs):
4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median, and Mode

- **Mean**

$$\frac{4+5+5+6+7+8+8+9+30}{9} = \frac{82}{9} = 9.11$$

- **Median**
    - Number of observations = 9
    - Middle (5th) value = **7**
- **Mode**
    - 5 and 8 occur twice.
    - **Mode = 5 and 8 (Bimodal)**

2. Range

$$30 - 4 = 26$$

Range = 26

(x)	(x-\bar{x})	((x-\bar{x})^2)
4	-5.11	26.12
5	-4.11	16.90
5	-4.11	16.90
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23

(x)	$(x-\bar{x})$	$((x-\bar{x})^2)$
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.23

Sum of squared deviations = **512.76**

Variance

$$\frac{512.76}{9} = 56.97$$

Standard Deviation

$$\sqrt{56.97} = 7.55$$

Variance ≈ 56.97
Standard Deviation ≈ 7.55

4. Outlier

The value **30** is much larger than the remaining salaries.

Outlier = 30 lakhs

5. Best Measure of Central Tendency

Median (7 lakhs) is the best measure because the outlier (30 lakhs) increases the mean (9.11 lakhs), making it less representative of the typical employee salary.

Case Study 2 — Website Response Time

Data (ms):
120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Mean and Median

Mean

$$\frac{120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300}{10}$$

$$= \frac{1505}{10} = 150.5$$

Mean = 150.5 ms

Median

Since there are 10 observations,

$$\frac{132 + 135}{2} = 133.5$$

Median = 133.5 ms

2. Range

$$300 - 120 = 180$$

Range = 180 ms

3. Variance and Standard Deviation (Population)

Variance

$$\approx 2526.65$$

Standard Deviation

$$\sqrt{2526.65} \approx 50.27$$

Variance ≈ 2526.65
Standard Deviation ≈ 50.27 ms

4. Skewness

There is one unusually high response time (300 ms).

The data is right-skewed (positively skewed).

5. Why is the Median Better?

The response time of **300 ms** is an outlier that increases the mean.

- Mean = **150.5 ms**
- Median = **133.5 ms**

The **median** better represents the typical website response time because it is not affected much by extreme values.

Case Study 3 — Student Performance

Marks:
42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2, and Q3

There are 12 observations.

Q2 (Median)

$$\frac{52 + 55}{2} = 53.5$$

Lower half:

42, 45, 48, 50, 52, 52

Q1

$$\frac{48 + 50}{2} = 49$$

Upper half:

55, 58, 60, 62, 65, 95

Q3

$$\frac{60 + 62}{2} = 61$$

Q1 = 49
Q2 = 53.5
Q3 = 61

2. IQR

$$61 - 49 = 12$$

IQR = 12

3. Outlier Using 1.5 × IQR Rule

Lower Limit

$$49 - 1.5(12) = 31$$

Upper Limit

$$61 + 1.5(12) = 79$$

Any value greater than **79** is an outlier.

Outlier = 95

4. Mean and Median

Mean

$$\frac{684}{12} = 57$$

Mean = 57

Median

$$\frac{52 + 55}{2} = 53.5$$

Median = 53.5

5. Effect of the Outlier on the Mean

The score **95** is much higher than the other marks.

- It increases the **mean** from the typical group of scores.
- The **median (53.5)** remains relatively unaffected.

Therefore, the **median** gives a better representation of the typical student's performance than the mean.

Case Study 4 — E-Commerce Orders

Data:
18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the Mean, Median, and Mode

Mean

Original amounts

Equal shares

2

4

7

$$\bar{x} = \frac{2+4+7}{3} = \frac{13}{3} = 4\ \frac{1}{3}$$

x̄ is the arithmetic mean (simple average)

$$x_1$$

$$x_1$$

$$x_2$$

$$x_2$$

$$x_3$$

$$x_3$$

$$\text{Mean} = \frac{18+20+22+22+24+25+26+28+30+35}{10}$$

$$= \frac{250}{10} = 25$$

Mean = 25

Median

There are 10 observations.

$$\text{Median} = \frac{24+25}{2} = 24.5$$

Median = 24.5

Mode

The value **22** occurs twice, while all others occur once.

Mode = 22

2. Find the Variance and Standard Deviation (Population)

x	x − Mean	(x − Mean)²
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Sum of squared deviations = **228**

Variance

$$\frac{228}{10} = 22.8$$

Standard Deviation

$$\sqrt{22.8} \approx 4.77$$

Variance = 22.8

Standard Deviation ≈ 4.77

3. Calculate the Coefficient of Variation (CV)

$$\mathrm{CV} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$
$$= \frac{4.77}{25} \times 100 = 19.08\%$$

Coefficient of Variation ≈ 19.08%

4. What Does the Standard Deviation Tell the Company?

The **standard deviation (4.77)** shows how much the daily orders vary from the average of **25 orders**. Since the value is relatively small, most daily orders are close to the mean, indicating fairly consistent order volumes.

5. Would You Describe the Order Data as Highly or Moderately Variable?

The **coefficient of variation is about 19.08%**, which indicates **moderate variability**. The daily orders fluctuate somewhat but remain reasonably consistent, with no extreme outliers.

Final Answers

Measure	Value
Mean	**25**
Median	**24.5**
Mode	**22**
Range	**17** (35 − 18)
Variance	**22.8**
Standard Deviation	**4.77**
Coefficient of Variation	**19.08%**

Measure	Value
Variability	**Moderately Variable**

Case Study 5 — Comparing Two Data Science Models

Prediction Errors

- **Model A:** 2, 3, 3, 4, 4, 5, 5, 6
- **Model B:** 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the Mean Error for Both Models

Model A

$$\text{Mean} = \frac{2+3+3+4+4+5+5+6}{8} = \frac{32}{8} = 4$$

Mean Error = 4

Model B

$$\text{Mean} = \frac{1+1+2+4+5+7+8+9}{8} = \frac{37}{8} = 4.625$$

Mean Error = 4.625

2. Calculate the Variance (Population)

Model A

x	x − Mean	(x − Mean)²
2	-2	4
3	-1	1
3	-1	1

x	x − Mean	(x − Mean)²
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Sum of squared deviations = **12**

$$\text{Variance} = \frac{12}{8} = 1.5$$

Variance = 1.5

Model B

x	x − Mean	(x − Mean)²
1	-3.625	13.14
1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

Sum of squared deviations = **69.88**

$$\text{Variance} = \frac{69.88}{8} \approx 8.73$$

Variance ≈ 8.73

3. Calculate the Standard Deviation

Model A

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

The points have mean μ = 5 and σ = 1.5.

$$\sigma$$

$$\sigma$$

Drag points

$$\sqrt{1.5} \approx 1.22$$

Standard Deviation ≈ 1.22

Model B

$$\sqrt{8.73} \approx 2.95$$

Standard Deviation ≈ 2.95

4. Which Model Has More Consistent Predictions?

A **smaller standard deviation** means the prediction errors are closer to the mean.

- **Model A:** SD ≈ **1.22**
- **Model B:** SD ≈ **2.95**

Answer: Model A has more consistent predictions because its prediction errors vary less.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better?

No. A lower mean error alone does not always mean a better model. A good model should have:

- A **low mean error** (high accuracy), and
- A **low standard deviation** (consistent predictions).

A model with a slightly higher mean error but much more consistent predictions may perform better in practice.

Measure	Model A	Model B
Mean Error	**4.00**	**4.625**
Variance	**1.50**	**8.73**
Standard Deviation	**1.22**	**2.95**
More Consistent Model	**Model A**	—
Better Model	**Model A**, because it has both a lower mean error and a lower standard deviation.	